CONFIDENTIAL

SCHOOL OF COMPUTING
Faculty of Engineering

UNIVERSITI TEKNOLOGI MALAYSIA

SKILL-BASED TEST 1
SET B

SEMESTER II 2019/2020

SUBJECT CODE	**: SECJ/SCSJ 1023**
SUBJECT NAME	**: PROGRAMMING TECHNIQUE II**
YEAR/COURSE	**: 1 (SECB/SECJ /SECP/SECR/SECV)** **2 (SCSR/SCSV)**
TIME	**: 2.45 pm – 4.45 pm (2 Hours)**
DATE/ DAY	**: 30th APRIL 2020 (THURSDAY)**

INSTRUCTIONS TO THE STUDENTS:

- Please read the ***General Guidelines for the Programming Technique II Test*** that is shared in WhatsApp's Group and/ or provided in UTM e-learning
- Read the problem and instructions carefully.
- You are given **TWO HOURS** to complete the test inclusive the submission of your program.

IMPORTANT NOTES:

- All the **COMMENT STATEMENTS** in the submitted program **WILL NOT BE EVALUATED**.

SUBMISSION PROCEDURE:

- Only the source code is required for the submission and the source code's file shall be named as follows: ***Name_ matricsNo_section.cpp*** (i.e. *AinaAli_A19EC018_01.cpp*).
- You do not need to compress the file.
- Submit the source code file (i.e., sbt1.cpp) via the **UTM's e-learning system**.

Problem **[100 Marks]**

Write a complete C++ program that contains a class called **Person**.

1. The class should have seven (7) **`private`** member variables (attributes) to hold a name, National Registration Identity Card (NRIC) number that has 12-digit numbers, height in centimeter (cm), weight in kilogram (kg), activity level, Basal Metabolic Rate (BMR), and calorie.

   Note: You must create the object of **`string`** class to store/ hold the NRIC number.

2. The class should have the following **`public`** member functions:

   - A **constructor with default arguments** that assign the values of name, height, weight and activity level member variables with the passed arguments. The constructor also enables to assign empty string ("") to the name and NRIC member variables, zero (0.0) to the height and weight member variables and zero (0) to the activity level member variable.

   - A **getName** function that returns the value of name member variable.

   - A **getGender** function that returns the string value of gender. The gender will be determined by the last digit of NRIC number. If the last digit is odd number, then the gender of the Person's object is male; otherwise gender of the Person's object is female.

     Example: *NRIC number is 980203041527. The last digit is 7. It is an odd number. Therefore, the gender of Person's object is male.*

   - A **getAge** function that returns the integer value of age. The age will be determined by the first two digits of NRIC number.

     Example: *NRIC number is 980203041527. The first two digits are 98, meaning the Person's object was born in 1998. His/ her age is 22.*

     *****Note:*** You must use an appropriate function(s) in **`string`** class to determine the age and gender of Person's object.

   - A **calcCalorie** function that calculates the value of BMR and calorie member variables with gender and age as the arguments.

     i) First step is to calculate the value of BMR variable according to the following *Mifflin – St Jeor* equation formula:

- Formula for male:

  BMR = (10 x weight) + (6.25 x height) – (5 x age) + 5,

  where weight is in kilogram (kg), height in centimeter (cm) and age in year.

- Formula for female:

  BMR = (10 x weight) + (6.25 x height) – (5 x age) - 161,

  where weight is in kilogram (kg), height in centimeter (cm) and age in year.

ii) Second step is to calculate the value of calorie variable from the BMR result in the first step and based on the activity level according to the following table:

Activity Level	**Formula**
Sedentary	Calorie = BMR x 1.2
Lightly Active	Calorie = BMR x 1.375
Moderately Active	Calorie = BMR x 1.55
Very Active	Calorie = BMR x 1.725
Extra Active	Calorie = BMR x 1.9

- A **readInput** function that reads the values from the keyboard for name, NRIC, height, weight and activity level member variables. For activity level, the input value is according to the following table:

Activity Level	**Input**
Sedentary	1
Lightly Active	2
Moderately Active	3
Very Active	4
Extra Active	5

*****Note:*** It is also need to invoke **getGender**, **getAge** and **calcCalorie** function to obtain the value of calorie.

- An **overloaded equality (==) operator** that determines if two objects have the same activity level, then return **true**; otherwise return **false**.

3. The class should also have the following **`friend`** of standalone functions:

- A **dispInfo** function to display the age and gender of Person's object in the following format (please refer to the sample output given in **Figure 1)**:

```
Your gender is ?????
Your age is ## year old
```

- An **overloaded output (<<) operator** function to display the object in the following format (please refer to the sample output given in **Figure 1**):

```
Hello, ????? ?????
Your gender is ????
Your age is ## years old
Your height is #.## cm
Your weight is #.## kg
Your BMR is #.## kcal/day
Your calorie based on activity level is #.## kcal/day
```

4. Write an appropriate **main** function that do the following tasks:

   - Create two **Person** objects named **p1** and **p2**. The objects have the following data:

Object	**p1**	**p2**
Name	""	Alisa Hashim
Height	0.0	150
Weight	0.0	50
Activity Level	0	2

   - Using an appropriate function(s), read the required input values and print the data of the appropriate object along with the notes about activity level for two objects (**p1** and **p2**) either both have same or different activity level.

Figure 1 shows a sample screen output that your program should produce. Note that, **bold** texts indicate keyboard input entered by the user. **Table 1** is the assessment criteria for the program.

```
Calorie Calculator

<<< Input >>>
Name: Arshad Amir
NRIC Number (without hyphen'-'): 001123055261
Height (in meter): 182
Weight (in kg): 75
Choose the activity level: 1-Sedentary, 2-Lightly Active, 3-Moderately Active,
4-Very Active, 5-Extra Active
Activity level (1 - 5): 1

<<< Output >>>
Hello, Arshad Amir
Your gender is male
Your age is 20 years old
Your height is 182.00 cm
Your weight is 75.00 kg
Your BMR is 1792.50 kcal/day
Your calorie based on activity level is 2151.00 kcal/day

**Note: Arshad Amir and Alisa Hashim have different activity level.
```

(a) Example of run 1

```
Calorie Calculator

<<< Input >>>
Name: Rafidah Rahman
NRIC Number (without hyphen'-'): 850711036032
Height (in meter): 154
Weight (in kg): 41.5
Choose the activity level: 1-Sedentary, 2-Lightly Active, 3-Moderately Active,
4-Very Active, 5-Extra Active
Activity level (1 - 5): 2

<<< Output >>>
Hello, Rafidah Rahman
Your gender is female
Your age is 35 years old
Your height is 154.00 cm
Your weight is 41.50 kg
Your BMR is 1041.50 kcal/day
Your calorie based on activity level is 1432.06 kcal/day

**Note: Rafidah Rahman and Alisa Hashim have same activity level.
```

(b) Example of run 2

Figure 1: Example runs of program

Table 1: Assessment criteria

Item	Criteria	Marks
A	i) The program is able to run.	5
	ii) Using an appropriate structure for the program - the code is properly indented - all the required header files are included - the main function is properly written including proper input and output formatting used	 1 1 2
B	The definition of class Person i) Data members ii) Constructor iii) getName function iv) getGender function v) getAge function vi) calcCalorie function vii) readInput function viii) Overloaded operator==	1 5 8 2 9 11 15 11 6
C	The definition of friend functions i) dispInfo function ii) Overloaded output (<<) operator	 3 10
D	Function main i) Object p1 and p2 creation ii) Read the required input values iii) Display the data of appropriate object iv) Display the notes about activity level for two objects	 3 1 1 5
Total		**100**